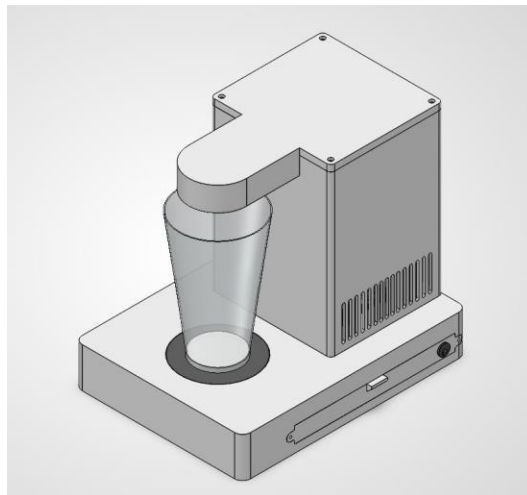




DiyaMithuru

PRODUCT CONTRACT



SMART LIQUID FILLING ASSISTANT

Product	DiyaMithuru
Category	Assistive Technology
Prototype	Alpha
Design focus	Inclusive, user-centered operation

1. PRODUCT DESCRIPTION

DiyaMithuru is an inclusive smart liquid filling assistant designed to reduce the need for visual judgement and manual pouring. The alpha prototype uses an ESP32-based control system, ultrasonic sensing for cup presence, temperature sensing, relay-controlled pumping and audio indication using web dashboard. When a cup is correctly positioned, the system can initiate controlled dispensing, monitor the liquid condition, provide visual and non-visual feedback using web site , and stop the pump according to the programmed filling condition.

ALPHA SCOPE

Prototype validation is based primarily on water and indoor home-use conditions. It is not a certified medical dosing device and should not be presented as one.

2. VALUE PROPOSITION

Who is the user?	Primary: people with full visual impairment who need a non-visual way to fill a cup. Secondary users include older adults, people with reduced dexterity, caregivers, and people recovering from illness who benefit from easier hydration routines.
How does the user benefit?	Less dependence on sight and continuous manual pouring; easier cup placement and filling; audio status feedback; temperature awareness and reduced risk of accidental dispensing when a cup is not correctly positioned.
Where and when is it used?	Primarily indoors at home, including kitchens, dining spaces, bedside areas and assisted-living environments. It can support routine drinking-water and beverage preparation.
Why would someone want it?	It combines simple automation and inclusive interaction in one affordable local product, helping turn a common daily task into a more predictable and independent experience.

3. PRIMARY USER PERSONA

Persona	Nadeesha - hypothetical primary user
Profile	32-year-old adult with full visual impairment who wants to prepare drinking water independently at home.
Current challenge	Cannot visually judge cup position or liquid level and may need assistance to avoid spills or unsafe handling.
Goals	Independent operation, minimal controls, clear audio feedback, predictable filling, simple cleaning, and safe use around the home.
Critical needs	Non-visual operation; cup-presence interlock; automatic pump control; temperature awareness; audible start/complete/warning feedback.

4. TARGET MARKET

Primary market: Individuals and households where a visually impaired person wants greater independence in daily liquid filling.

Secondary market: Older adults, people with reduced hand dexterity or mobility, caregivers, and families supporting children or people in recovery.

Institutional market: Rehabilitation centers, elder-care settings, disability-support organizations, selected healthcare environments, NGOs and inclusive-living programs.

Positioning: Affordable, locally developed assistive technology focused on simple operation, accessibility, and everyday practicality.

5. USER NEEDS, PRODUCT ATTRIBUTES & ENGINEERING SPECIFICATIONS

The following requirements convert user needs into measurable design targets. Numeric values are acceptance targets for the alpha prototype and should be confirmed by testing before final submission.

User Need	Product Attribute	Required Function	Engineering Specification / Acceptance Target	Verification
Independent, non-visual use	Cup detection	System shall identify a correctly positioned cup before enabling liquid dispensing.	Target: $\geq 95\%$ successful detection in 20 placement trials.	Repeated placement test
No accidental dispensing	Safety interlock	Pump shall remain OFF when no cup is detected.	Target: 20/20 no-cup trials with zero unintended pump activation.	No-cup safety test
Quick operation	Control response	System should respond promptly after valid cup detection.	Target: pump control response ≤ 1.0 s after stable valid detection.	Stopwatch
Consistent filling	Automatic dispensing	Liquid shall be dispensed using a programmed filling quantity rather than under the same cup, pump continuous manual and supply conditions.	Target: repeated fill quantity within $\pm 10\%$	10-fill volume test
Automatic completion	Pump control	Pump shall turn OFF automatically at the programmed fill every cycle; maximum condition and include a continuous pump ON time maximum run-time ≤ 60 s. safeguard.	Target: automatic stop	Cycle test
Temperature awareness	Temperature monitoring	System shall measure liquid temperature over the normal beverage range.	Design target: 0-100 degC measurement range; ± 2 degC accuracy from 20-80 degC.	Compare reference thermometer with

User Need	Product Attribute	Required Function	Engineering Specification / Acceptance Target	Verification
Clear non-visual feedback	Audio indication	Provide audible indication for key states such as ready/start, completion and warning.	Target: distinct audible cues; minimum 65 dBA at 0.5 m in a quiet indoor setting.	Sound-level / user recognition test
Simple interaction	User interface	Normal operation shall not require a user action after placing touchscreen or visual menu.	Target: ≤ 1 intentional user action after placing the cup; audio feedback available.	Task analysis / user trial
Low-voltage home use	Power & electrical safety	Use a low-voltage DC supply for the prototype electronics and pump system.	5-12 V DC system supply; wall adapter used as external mains interface.	Visual inspection / voltage check
Reliable operation	System reliability	System shall complete repeated fill cycles without controller lock-up or pump-control failure.	Target: ≥ 100 consecutive bench cycles without control failure.	Endurance test
Indoor liquid handling	Operating scope	Alpha prototype shall be validated with water and similar low-viscosity test liquids only after compatibility check.	Minimum validation medium: water; electronics protected from normal splash exposure by enclosure/layout.	Inspection / functional test
Affordable ownership	Cost target	Product should remain accessible relative to complex imported assistive systems.	Planning target: manufacturing cost $\leq$ LKR 9,500; selling price $\leq$ LKR 15,000.	BOM and cost review

6. ALPHA PROTOTYPE ACCEPTANCE CRITERIA

The alpha prototype will be considered successful when the core assistive function can be demonstrated safely and repeatedly under the defined test conditions.

- Cup presence is detected reliably before dispensing begins.
- The pump does not activate when a cup is absent or outside the valid placement zone.
- Dispensing starts and stops automatically according to the programmed control logic.
- Temperature sensing produces stable readings that are reasonably consistent with a reference thermometer.
- Audio indication communicates key system states without requiring visual feedback.
- The device completes repeated demonstrations without leakage reaching exposed electronics or causing control failure.
- The primary user or a suitable representative can understand the operating sequence with minimal instruction.
- The final prototype enclosure and product form present a resolved, natural-product appearance suitable for exhibition demonstration.

7. Verification Plan

Test	Method	Pass Evidence
Cup detection	20 correct cup placements	Detection success rate and false negatives
No-cup interlock	20 trials without a cup	Unintended pump activations = 0
Fill repeatability	10 fills using same cup and setup	Measure each fill volume; calculate variation
Automatic stop	10 full cycles	Pump stops without manual continuous control
Temperature	Compare at approximately 20, 40, 60 and 80 degC	Error against reference thermometer
Audio indication	Test ready/start/complete/warning cues	Cue is audible and distinguishable at 0.5 m
Endurance	100 bench cycles	Record control failures, leaks or abnormal operation
User validation	Task trial with target-user representative	Observe independence, errors, ease of use and feedback

8.PRODUCT BOUNDARIES & ASSUMPTIONS

- The alpha prototype is intended for indoor demonstration and home-assistance scenarios.
- The product is not positioned as a certified medical device, clinical fluid meter, or replacement for professional medical advice.
- For the final engineering report, measured prototype dimensions, weight, actual sensor accuracy, measured fill repeatability, and endurance results should replace any unverified target values.
- Web-based logging, mobile applications, AI recognition and smart-home connectivity are outside the current alpha-prototype contract unless separately implemented and demonstrated.
- Commercial claims such as warranty, delivery, manufacturing cost and retail price are business-plan assumptions until a real production and service process is established.

9.CONTRACT SIGN-OFF

Prepared by

Reviewed by

Date